

Ronak Chavva

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Education

University of Texas at Dallas

May 2027

Bachelor of Science in Computer Science

GPA: 4.0 / 4.0

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Linear Algebra, Statistics

Projects

Binary Search Gang (BSG)

Node.js, PostgreSQL, Redis, Kafka, WebSockets

- Built a **stateful, distributed backend** handling user sessions, rooms, live submissions, and leaderboards with support for concurrent users across isolated game instances.
- Designed normalized **PostgreSQL schemas** for relational data and reduced read latency by **20%** through targeted **Redis caching** of hot query paths.
- Implemented a **WebSocket** service for bidirectional real time communication to synchronize live room state across all connected clients.
- Engineered **event driven workflows with Kafka** to propagate state updates between decoupled services, improving throughput and reducing interservice coupling.

Firewall Management Dashboard

React, TypeScript, AWS Lambda, API Gateway, S3, Terraform

- Built a **serverless production platform** to visualize and manage enterprise firewall rules, replacing a brittle legacy API with a clean, maintainable service layer.
- Implemented backend logic using **AWS Lambda** exposed via **API Gateway** and **Route 53**, with all infrastructure defined in **Terraform**.
- Designed a **React + TypeScript** frontend enabling non technical stakeholders to query and audit network policy data without direct API access.

Experience

State Farm

Richardson, TX

Software Engineer

2023 - Present

- Developed and maintained a **production full stack platform** using **React, TypeScript, and Terraform** to manage enterprise firewall configurations serving internal engineering teams.
- Facilitated a **\$400M infrastructure migration** from TP2 to **AWS ROSA**, validating cloud resource integrity and end to end fidelity across distributed environments.
- Automated repetitive manual workflows via scripting, reducing engineering overhead on non critical processes by **80%** and freeing capacity for feature work.

ACM - Binary Search Gang (BSG)

Richardson, TX

Lead Developer

2025 - Present

- Architected and led development of a **Chrome extension** integrating real time party chat and a statistics observer into LeetCode, with a roadmap toward leaderboards and ranked play.
- Designed the **end to end system architecture**, extension messaging, backend services, and data persistence, making foundational decisions that established scalable, maintainable patterns for the team.
- Diagnosed and resolved **deeply rooted tech debt** inherited from prior contributors, reverse engineering undocumented logic and refactoring core systems without regressions.
- Debugged complex **race conditions and state synchronization failures** across the extension/backend boundary, unblocking feature development across multiple sprints.

Skills

Languages: C, C++, C#, Go, Python, Java, TypeScript, JavaScript, SQL, Bash

Backend & Databases: PostgreSQL, Redis, Kafka, Node.js, WebSockets, REST, MongoDB

Cloud & Infrastructure: AWS (ROSA, S3, Lambda, API Gateway, Route 53), Terraform, Docker, Kubernetes, CI/CD

Frontend & Tools: React, Angular, Git, Postman, Docker Compose, Terraform CLI, AWS CLI